



POS M Packet Template

Professional template for onboarding packets, design notes, and project briefs

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Draft date: June 20, 2026

Project mission. Provide a consistent structure for POS M project documentation. Each packet should explain what is being built, why it matters, what assumptions are frozen, what interfaces exist, what counts as done, and how a new contributor should start useful work.

Template baseline. Use the black-background POS M wordmark on the cover, keep section headings black, place a very small POS M mark in the footer on every page, and use restrained accent color only in note boxes, warnings, links, and small rules.

How to read this packet. Sections 1–3 define the purpose, scope, and system view. Sections 4–7 define requirements, interfaces, implementation, and verification. Sections 8–10 provide acceptance criteria, repository structure, and resources. Replace placeholder text before publishing a real packet.

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1 Start Here: What Are We Building?

This packet is a reusable POSM documentation template. It is meant for onboarding packets, design briefs, correction notes, and technical handoffs. It follows the same restrained packet language used in previous POSM documents: a short mission statement, frozen baseline assumptions, tables for decisions, and clear acceptance criteria.

1.1 The one-sentence version

We are creating a standard packet format that lets POSM produce clean, consistent project documents without redesigning the layout every time.

1.2 Why this matters

POSM projects move faster when the design context is written down. A good packet should answer the questions that usually slow a project down: what is the goal, what is frozen, what is still unknown, what interface is being built, and what would count as a real first handoff?

POSM note

Keep the tone technical and direct. The packet should be readable by a smart student seeing the project for the first time, but it should not sound like a brochure or a marketing sheet.

1.3 What this template does and does not do

Does	Does not
Gives a consistent packet structure.	Replace project-specific engineering judgment.
Provides title, contents, note boxes, tables, code blocks, and checklist styling.	Force every POSM project into the exact same section order.
Keeps branding visible but restrained.	Turn packets into graphic-design posters.
Supports Overleaf and local LaTeX workflows.	Require a custom build system.

2 System-Level Decision Summary

Use this table to collect frozen assumptions and intentional boundaries. When a decision is not frozen, label it as preliminary.

Topic	Decision
Visual style	Classic technical packet: serif typography, black headings, thin rules, clean tables, restrained note boxes.
Cover logo	Full POSM wordmark with black background, larger than the body-page mark but not oversized.
Body-page logo	Tiny dotted POSM mark in the lower-left footer on every page.
Accent color	Used sparingly in note boxes, important boxes, links, and small rules only.
Default title color	Black. Do not use blue section headings.
Compilation target	pdfLaTeX on Overleaf or local TeX Live.

3 Packet Structure

3.1 Recommended sections

For most POSM onboarding packets, start with the following structure and delete what is not needed.



1. Start Here: what problem are we solving?
2. System-level decision summary.
3. Theory needed for this project.
4. System overview and block diagram.
5. Hardware, software, or experimental interfaces.
6. Implementation plan.
7. Verification and validation plan.
8. Acceptance checklist.
9. Repository structure.
10. Resources and reading map.

Important

The packet should make unknowns visible. If a design value is estimated, preliminary, or awaiting measurement, say that directly instead of hiding it in polished wording.

3.2 Plain block diagram style

Use a simple text diagram when a formal figure would add friction but not clarity.

```
physical input / user command
    |
    v
front-end or measurement block
    |
    v
processing / controller / estimator
    |
    v
output interface / actuator / data product
    |
    v
experiment, user, or downstream POSM system
```

4 Interfaces

Use exact names wherever possible: connector pins, file names, register names, part numbers, function names, optical ports, or mechanical datums.

Name	Direction / type	Notes
INPUT_A	Input	Replace with range, impedance, units, protocol, or expected behavior.
OUTPUT_A	Output	Replace with output range, load, safety limit, file format, or timing.
STATUS_A	Status	Replace with fault, saturation, health, ready, locked, calibrated, or similar status behavior.

5 Implementation Plan

A useful first handoff is the smallest version that proves the core idea. Avoid starting with final polish.



Phase	Goal	Exit criterion
0	Freeze assumptions and collect references.	Assumptions table reviewed.
1	Build a model, mockup, or minimal signal path.	Core data or signal path works.
2	Integrate the real interface.	Real part, hardware, file, or experiment communicates.
3	Measure performance and document limits.	Plots, data, and acceptance checklist complete.

6 Verification and Test Plan

Testing should be designed before the final build. This prevents “it works” from meaning “we saw one encouraging screenshot.”

Test	Method	Pass condition
Basic function	Stimulate the system with a known input.	Expected output appears with correct sign, scale, or timing.
Range / limit	Sweep the input or operating range.	No unsafe behavior; saturation and failure points are known.
Noise / drift	Record over time.	Noise floor or drift is below the requirement or honestly documented.
Fault behavior	Inject missing input, invalid state, rail, or overrange.	Safe response and clear status.

7 What Counts as Done

A serious first handoff should satisfy the following:

- README explains the project, system boundary, and how to run the first demo.
- Assumptions are separated into frozen, estimated, and unknown.
- Interfaces are named exactly and documented in one place.
- Calculations or simulations reproduce the chosen design values.
- The first measurement or demo produces reviewable data, not just a verbal report.
- Known limitations and next steps are documented honestly.

8 Suggested Repository Structure

```
posm_example_project/
  README.md
  docs/
    00_project_brief.md
    01_theory_and_assumptions.md
    02_interfaces.md
    03_verification_plan.md
    04_reading_list.md
  hardware/
    schematic/
    layout/
    datasheets/
    calculations/
  software/
```



```
src/
tests/
scripts/
mechanical/
  cad/
  drawings/
outputs/
  data/
  plots/
  reports/
```

9 Resources and Reading Map

Each entry should say why the resource matters. A link dump is less useful than a map.

Topic	Resource	Use it for
Core concept	Textbook, paper, internal packet, or application note.	Extract the physical model, design rule, or accepted method.
Part / module	Datasheet or official documentation.	Pin names, timing, limits, dimensions, and examples.
Implementation	Reference design, tutorial, or previous POSM repo.	Useful implementation ideas; do not copy blindly.

10 Final Notes

The goal is not to make the packet look fancy. The goal is to make the project executable: clear enough that a member can start, disciplined enough that the design does not drift, and honest enough that unknowns are visible.

